

Install & run guide

devopsiq — from zero to investigating real DevOps problems

Package	devopsiq (PyPI) · console command: devopsiq
Docker image	ghcr.io/devopsabhii/devops-ai-agent (public, amd64 + arm64)
Requires	Python 3.10+ (or just Docker) · MIT license
Key needed?	No — the agent runs keyless for record commands and the tool layer; only model questions need an OpenRouter API key
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1. Install — pick one of three ways

1.1 From PyPI (fastest, no clone)

Requirement: Python 3.10+. pipx: pip install pipx if you don't have it.

```
pipx install devopsiq          # recommended (isolated tool)
# or: pip install devopsiq
```

```
# check it works — no key needed for this:
devopsiq /report
# → "(no investigation recorded)", exit 0
```

1.2 Docker (everything bundled — CLIs included)

The image bundles kubectl, helm, gh, trivy, git, curl, docker CLI.

```
docker run --rm -it \
-e OPENROUTER_API_KEY=sk-or-... \
-v "$HOME/.kube:/home/agent/.kube:ro" \
-v agent-records:/data \
ghcr.io/devopsabhii/devops-ai-agent --json "check the cluster"
```

- The `-v "$HOME/.kube"` mount gives the agent your Kubernetes context (read-only). Drop it if you only use Docker tools.
- Add `-v /var/run/docker.sock:/var/run/docker.sock` to let the Docker/Compose tools talk to your local daemon.
- `-v agent-records:/data` keeps investigation records between runs (the image sets `AGENT_STORE_DIR=/data`).
- Investigations are read-only by construction: every tool is a fixed, allowlisted command template — nothing in the agent can mutate your cluster or files.

1.3 From source (development)

Every devopsiq command below maps 1:1 to `.venv/bin/python main.py`.

```
git clone https://github.com/DevOpsAbhii/devops-ai-agent.git
cd devops-ai-agent
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt

.venv/bin/python main.py /report    # same console as devopsiq
```

1.4 The CLIs of the domains you use

The agent shells out to the real CLIs — install only the ones you need; each tool reports its exact error if its CLI is missing, so a partial install is fine: kubectl (Kubernetes), docker + docker compose (containers), terraform / helm / argocd / istioctl / trivy (infra & security), git and gh (repos & Actions), aws / gcloud / az (cloud identity), curl (monitoring endpoints), ansible-inventory / ansible-playbook (Ansible).

2. Set up credentials (or none at all)

2.1 With an OpenRouter API key (full agent)

Get a key at openrouter.ai/keys. Never commit the key anywhere.

```
# option A – a .env file next to the install (or in the repo for source installs):
cp .env.example .env 2>&/dev/null || true
echo 'OPENROUTER_API_KEY=sk-or-v1-...' && .env
```

```
# option B – your shell:
export OPENROUTER_API_KEY=sk-or-v1-...
```

2.2 No key? The agent still runs (model-less mode)

- The REPL starts without a key: it prints a [setup] notice and continues.
- Record commands work fully: /investigate, /investigation, /investigations, /report, /endinvestigation.
- The entire 58-tool library works (as a Python package or via the investigation records).
- Only questions to the model exit 1 with a loud setup message — you never get a silent failure or an invented answer.

2.3 No OpenRouter? Point it at anything OpenAI-compatible

Same client, same agent – only the endpoint changes.

```
# local Ollama – free and offline:
export OPENROUTER_BASE_URL=http://localhost:11434/v1
export OPENROUTER_MODEL=llama3.2          # any Ollama model that does tools
devopsiq "docker container checkout keeps exiting, investigate"
```

3. Choose your model

The default is z-ai/glm-5.3 — a cheap, reliable tool-caller (the agent is a tool-use loop, so pick a model with solid function calling; an investigation makes several calls per run, so price-per-call multiplies). Four ways to change it — the first one set wins:

Priority	How	Scope
1	--model NAME flag	One run only
2	/model <name> (REPL command)	Saves to ~/.devops-ai-agent/config.json — every future run
3	OPENROUTER_MODEL env var (or .env)	Every run on this machine
4	built-in default (z-ai/glm-5.3)	Fallback

Transient failures (timeouts, 429s, 5xx) retry automatically with backoff.

per-run override:

```
devopsiq --model openai/gpt-5.2 "why is api-5d6f crash-looping?"
```

persistent default (from the REPL):

```
You: /model openai/gpt-5.2
```

```
Model set to openai/gpt-5.2 (saved in /home/you/.devops-ai-agent/config.json).
```

show the current model any time:

```
You: /model
```

```
Model: openai/gpt-5.2
```

```
Config file: /home/you/.devops-ai-agent/config.json (save a default with /model &lt;name&gt;)
```

4. Run it

4.1 Interactive REPL

Banner shows the model, backend, store location, and 58 tools.

```
devopsiq
```

Slash commands are handled locally – they never reach the model.

```
DevOps AI Agent (Phase 9 – 58 read-only tools, persistent investigations)
Model:    z-ai/glm-5.3
Backend:  https://openrouter.ai/api/v1
Store:    /home/you/.devops-ai-agent/investigations
```

You: The checkout service container keeps exiting in Docker. Investigate.

Agent: (docker_ps → docker_inspect → docker_logs, tracks hypotheses, ends with the structured report: crash command, exit code, remediation)

You: /report ← canonical report from the record, any time

You: exit

Command	What it does
/investigate <problem>	Open a formal investigation (same path the model uses)
/investigation	Show the live tracked state (hypotheses/evidence) · alias /status
/investigations	List saved records on disk, newest first · alias /history
/report	Show the canonical report (once concluded)
/endinvestigation	Clear the record (the saved copy stays as history) · alias /end
/model [<name>]	Show the active model, or save a default and switch to it

4.2 One-shot (cron / CI / scripts)

Exit codes: 0 = completed, 1 = setup/API error – safe for pipelines.

```
devopsiq "why is api-5d6f crash-looping?"          # investigate, print, exit
devopsiq --json "why is api-5d6f crash-looping?"    # structured JSON report
devopsiq --resume --json "any update?"              # continue a prior run
devopsiq --store-dir /tmp/runs --out report.json "..."/>
devopsiq /report                                     # slash commands work one-shot
```

- --json prints one JSON document: problem, status, hypotheses, evidence, and conclusion (root_cause, remediation, verification, confidence) — a pipeline can act on the verdict.
- --out PATH writes that JSON to an exact file; --store-dir DIR redirects persistence for the run; --resume continues the newest in-progress record.
- Records auto-save to ~/.devops-ai-agent/investigations/ on every change — the REPL resumes the newest in-progress one at startup.

4.3 Verify your install in 30 seconds

If the first line works, the install is good.

```
devopsiq /report                                     # works with NO key → "(no investigation
    recorded)"
devopsiq /model                                       # shows the active model
devopsiq "what files changed in the last commit?"    # needs the key + git installed
```

5. When something doesn't work

Symptom	Meaning	Fix
[input] OPENROUTER_API_KEY is not set (exit 1 on a question)	Model-less mode: you asked the model something with no key	Add the key (section 2.1) or use /report-style commands, which work keyless
[input] OPENROUTER_API_KEY still has the placeholder value	The .env still says your_key_here	Replace your_key_here with the real sk-or-v1-... key
[error] The API key was rejected (HTTP 401)	The key is wrong, expired, or revoked	Re-check the key at openrouter.ai/keys; mind shell exports override .env
[error] The OpenRouter API returned an error (HTTP 4xx/5xx) after several seconds	Transient failures are retried 3× with backoff before this message appears	Retry later; persistent 5xx = OpenRouter-side issue
Tool error: k8s_get...: kubectl: command not found	That domain's CLI isn't installed	Install the CLI (section 1.4) — or ignore if you don't use that domain
Every k8s tool errors with connection refused	No cluster reachable from your kubeconfig	Point ~/.kube/config at a cluster, or mount it in Docker
pip installs an older version than the latest	pip's local HTTP cache	pip install --no-cache-dir --upgrade devopsiq

What the agent will never do: it cannot mutate anything. All 58 tools are read-only by construction — fixed command templates, validated names, no shell, no writes to your cluster or files. Investigation records live only in ~/.devops-ai-agent/.

Install & run guide — generated from the repository on 27 September 2026. Regenerate: .venv/bin/python docs/generate_user_guide.py